INCH-POUND

AN916 Rev 11
19 July 2011
SUPERSEDING
AN916 Rev 10
10 February 1984

DETAIL SPECIFICATION SHEET

ELBOW, PIPE, INTERNAL THREAD, 90°

Reinstated after 19 July 2011. Inactive for new design.
For new design, use SAE-AS4856.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4842.

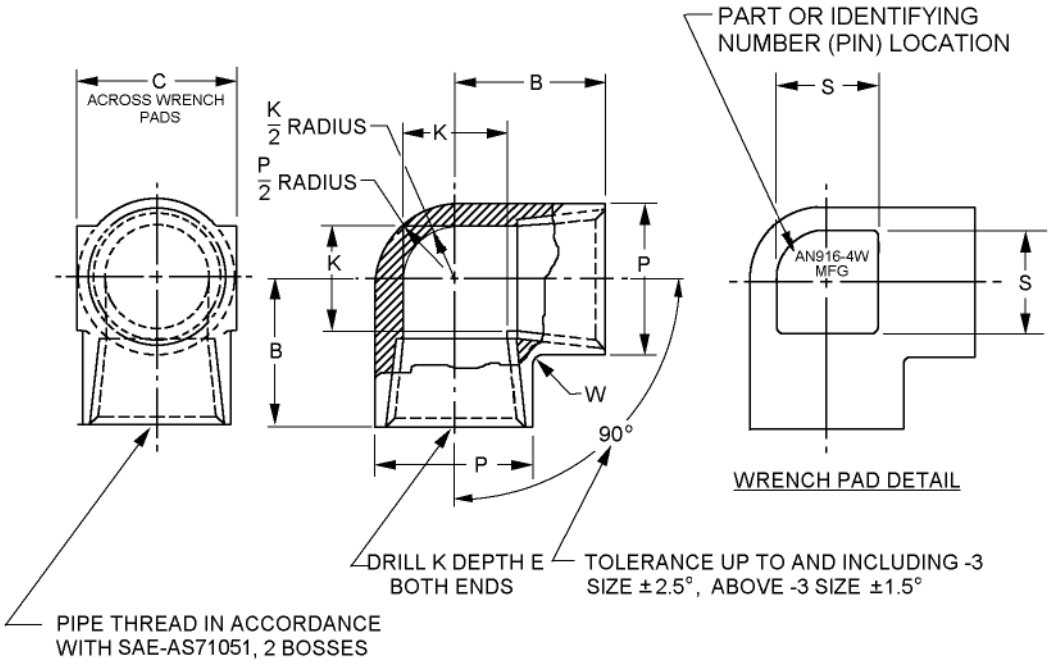


FIGURE 1. Elbow 90° dimensions and configuration.

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Dash number	A Pipe thread ANPT SAE-AS71051	B +.046 (1.17) -0.00 inches (mm)	C inches (mm)	E +.046 (1.17) -0.00 inches (mm)
1	1/8-27	.641 (16.28)	.625 (15.88)	.641 (16.28)
2	1/4-18	.891 (22.63)	.813 (20.65)	.953 (24.21)
3	3/8-18	.969 (24.61)	.938 (23.83)	1.094 (27.79
4	1/2-14	1.234 (31.34)	1.188 (30.18)	1.359 (34.52)
6	3/4-14	1.344 (34.14)	1.375 (34.93)	1.469 (37.31)
8	1-11 1/2	1.625 (41.28)	1.750 (44.45)	1.813 (46.05)
10	1/ 1/4-11 1/2	1.688 (42.88)	2.156 (54.76)	1.875 (47.63)

Dash number	K Dia. inches (mm)		P Dia. inches (mm)	S Approx inches (mm)	W Rad. inches (mm)
1	.327 (8.31)	+.000 -.016 (0.41)	.578 (14.68)	.500 (12.70)	.063 (1.60)
2	.423 (10.74)	+.000 -.019 (0.48)	.781 (19.84)	.625 (15.88)	.094 (2.39)
3	.557 (14.15)	+.000 -.028 (0.71)	.922 (23.42)	.750 (19.05)	.094 (2.39)
4	.688 (17.48)	+.000 -.034 (0.86)	1.156 (29.36)	.875 (22.23)	.125 (3.18)
6	.897 (22.78)	+.000 -.040 (1.02)	1.359 (34.52)	1.000 (25.40)	.125 (3.18)
8	1.128 (28.65)	+.000 -.037 (0.94)	1.688 (42.88)	1.125 (28.58)	.125 (3.18)
10	1.471 (37.36)	+.000 -.036 (0.91)	2.094 (53.19)	1.250 (31.75)	.156 (3.96)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.015 inch (0.38 mm).
4. Break sharp edges .and remove all hanging burrs and slivers
5. Machined surfaces shall be finished to 125µin Ra, forged surfaces shall be 250µin Ra, unless otherwise specified on the figures. Surface finish shall be in accordance with ASME B46.1.
6. For design features purposes, this standard takes precedence over documents referenced herein.
7. Referenced documents shall be of the issue in effect on date of invitation for bid.

FIGURE 1. Elbow 90° dimensions and configuration - Continued.

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REQUIREMENTS:

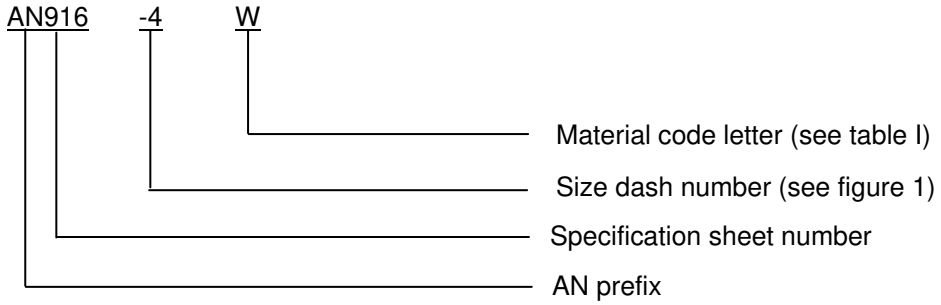
Dimensions and configuration shall be in accordance with figure 1.

Materials and finishes shall be in accordance with SAE-AS4842, see table I for material code.

TABLE I. Material and designators.

Code letter	Material
Blank	Copper alloy
J	Corrosion resistant steel (CRES), type 304
K	CRES, type 316Cannon
R	CRES. Type 321
W	Aluminum alloy 7075-T73

PIN. The PIN consists of the prefix “AN” the specification sheet number, a dash number for pipe size, and material code letter. Unassigned PIN’s shall not be used.



PIN example: AN916-4W indicates a 90° elbow, 1/2-14 ANPT internal pipe threads, aluminum alloy 7075-T73.

Supersession data:

Due to stress corrosion cracking aluminum alloys 2014 and 2024 “D” designator has been replaced by aluminum alloy 7075 “W” designator. Example: AN916-8D use AN916-8W.

Metal cracking due to high temperatures CRES alloy 347 “S” designator has been replaced by CRES alloy 321 “R” designator. Example: AN918-6S use AN916-8R.

Marking: Part shall be permanently marked with the AN PIN, and include the manufacturers CAGE, name, or trademark.

Table II provides a detailed cross-reference of AN916 PINs and replacement SAE-AS4856 PINs. Users are cautioned to evaluate replacements for their particular application.

CAUTION: The superseding information is valid as of the date of this specification and may be superseded by subsequent revisions of the superseding document.

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TABLE II. Cross-reference data.

AN PIN (inactive)	Pipe size	Replacement AS PIN (for new design)	Replacement AN PIN (inactive)
AN916-1	.125	AS4856-01	
AN916-1D	.125	AS4856W01	AN916-1W
AN916-1J	.125	AS4856J01	
AN916-1K	.125	AS4856K01	
AN916-1R	.125	AS4856R01	
AN916-1S	.125	AS4856R01	AN916-1R
AN916-1W	.125	AS4856W01	
AN916-2	.250	AS4856-02	
AN916-2D	.250	AS4856W02	AN916-2W
AN916-2J	.250	AS4856J02	
AN916-2K	.250	AS4856K02	
AN916-2R	.250	AS4856R02	
AN916-2S	.250	AS4856R02	AN916-2R
AN916-2W	.250	AS4856W02	
AN916-3	.375	AS4856-03	
AN916-3D	.375	AS4856W03	AN916-3W
AN916-3J	.375	AS4856J03	
AN916-3K	.375	AS4856K03	
AN916-3R	.375	AS4856R03	
AN916-3S	.375	AS4856R03	AN916-3R
AN916-3W	.375	AS4856W03	
AN916-4	.500	AS4856-04	
AN916-4D	.500	AS4856W04	AN916-4W
AN916-4J	.500	AS4856J04	
AN916-4K	.500	AS4856K04	
AN916-4R	.500	AS4856R04	
AN916-4S	.500	AS4856R04	AN916-4R
AN916-4W	.500	AS4856W04	

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TABLE II. Cross-reference data - Continued.

AN PIN (inactive)	Pipe size	Replacement AS PIN (for new design)	Replacement AN PIN (inactive)
AN916-6	.750	AS4856-06	
AN916-6D	.750	AS4856W06	AN916-10W
AN916-6J	.750	AS4856J06	
AN916-6K	.750	AS4856K06	
AN916-6R	.750	AS4856R06	
AN916-6S	.750	AS4856R06	AN916-10R
AN916-6W	.750	AS4856W06	
AN916-10	1.000	AS4856-08	
AN916-8D	1.000	AS4856W08	AN916-10W
AN916-8J	1.000	AS4856J08	
AN916-8K	1.000	AS4856K08	
AN916-8R	1.000	AS4856R08	
AN916-8S	1.000	AS4856R08	AN916-10R
AN916-8W	1.000	AS4856W08	
AN916-10	1.250	AS4856-10	
AN916-10D	1.250	AS4856W10	AN916-10W
AN916-10J	1.250	AS4856J10	
AN916-10K	1.250	AS4856K10	
AN916-10R	1.250	AS4856R10	
AN916-10S	1.250	AS4856R10	AN916-10R
AN916-10W	1.250	AS4856W10	

Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to SAE-AS4842, this document references the following:

ASME B46.1
SAE-AS4856
SAE-AS71051

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2011-077)

Review activities:
Navy - MC, SH
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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